Exercise 11.1

Consider the following controllable and observable system

$$\begin{cases} \dot{x} = \begin{bmatrix} -1 & 1 \\ -1 & -1 \end{bmatrix} x + \begin{bmatrix} 1 \\ 1 \end{bmatrix} u \\ y = \begin{bmatrix} 1 & 1 \end{bmatrix} x, \end{cases}$$

and associated full-order observer with observer gain

$$L = \begin{bmatrix} 2 \\ -9 \end{bmatrix}.$$

1. Set up the extended state matrices for the system

$$A_{\mathrm{e}} = \begin{bmatrix} A & 0 \\ C & 0 \end{bmatrix} B_{\mathrm{e}} = \begin{bmatrix} B \\ 0 \end{bmatrix}.$$

2. Find a feedback for the extended system, assigning poles of $A_{\mathrm{e}} + B_{\mathrm{e}}F_{\mathrm{e}}$ in $-1, -2, -3$. Use the command `place`.
3. Extract the feedback gain F and the integral gain F_I from the designed control F_{e}.
4. Compute the closed loop system

$$\begin{cases} \dot{x}_{\mathrm{cl}} = A_{\mathrm{cl}} x_{\mathrm{cl}} + B_{\mathrm{cl}} r \\ y = C_{\mathrm{cl}} x_{\mathrm{cl}} + D_{\mathrm{cl}} r, \end{cases}$$

   with matrices given by

$$\begin{aligned} A_{\mathrm{cl}} &= \begin{bmatrix} A & BF & BF_I \\ -LC & A + BF + LC & BF_I \\ C & 0 & 0 \end{bmatrix} \\ B_{\mathrm{cl}} &= \begin{bmatrix} 0 \\ 0 \\ -1 \end{bmatrix} \\ C_{\mathrm{cl}} &= \begin{bmatrix} C & 0 & 0 \end{bmatrix} \\ D_{\mathrm{cl}} &= 0. \end{aligned}$$

5. Verify the control by computing a step response, and see that the output converges to the reference.

Exercise 11.2

Consider the following controllable system

$$\begin{cases} \dot{x} = \begin{bmatrix} -1 & 1 \\ -1 & -1 \end{bmatrix} x + \begin{bmatrix} 1 \\ 1 \end{bmatrix} u \\ y = \begin{bmatrix} 1 & 1 \end{bmatrix} x. \end{cases}$$

1. Compute an optimal state feedback for this cost functional

$$\mathcal{J} = \int_0^\infty (\rho y^T y + u^T u) dt$$

   for $\rho = 1, 10, 100$. How does this change the eigenvalues of $A + BF$, and why?
2. Use Bryson's rule for choosing R and Q in the cost functional

$$\mathcal{J} = \int_0^\infty (x^T Q x + u^T R u) dt,$$

   when the maximal acceptable values of x_1^2 is 1, the maximal acceptable values of x_2^2 is 2, and the maximal acceptable values of u^2 is 3.
3. Simulate the system, and determine if the control performs as expected.